

Understanding Cap and Trade Systems: An Interactive Simulation Guide

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This guide introduces students to tradable emission permits through an interactive simulation. You will learn the history, theory, and practical application of cap-and-trade systems—one of the most powerful market-based tools for environmental regulation.

The Origins of Tradable Permits

The concept of tradable pollution permits emerged from economic theory in the 1960s.¹ John Dales first proposed the idea that pollution rights could be bought and sold like any other commodity, creating incentives for efficient pollution reduction.

¹ Dales, J.H. (1968). *Pollution, Property & Prices*. University of Toronto Press.

Why Tradable Permits?

Traditional environmental regulation often uses *command-and-control* approaches²—telling each polluter exactly how much to reduce. This is inefficient because:

² Command-and-control regulations specify exactly how much each polluter must reduce emissions, regardless of the cost.

- Different firms face different costs of reducing pollution
- A firm that can reduce cheaply should reduce more
- A firm facing high reduction costs should reduce less
- Markets can find this efficient allocation automatically

The tradable permit system harnesses market forces to achieve environmental goals at the lowest possible cost to society.

The key insight: Let those who can reduce pollution cheaply do more of the reduction, while those facing high costs can buy permits instead.

Real-World Applications

The U.S. Acid Rain Program (1990)

The most celebrated success story is the U.S. SO₂ trading program under the Clean Air Act Amendments of 1990.³

- **Goal:** Reduce sulfur dioxide emissions by 50%
- **Result:** Goal achieved two years early
- **Cost:** One-third of initial estimates
- **Innovation:** Spurred development of scrubber technology

³ Schmalensee, R., & Stavins, R.N. (2013). The SO₂ Allowance Trading System: The Irony History of a Grand Policy Experiment. *Journal of Economic Perspectives*, 27(1), 103-122.

The European Union Emissions Trading System

Launched in 2005, the EU ETS is the world's largest carbon market, covering over 11,000 installations.⁴

⁴ As of 2024, the EU ETS covers approximately 40% of the EU's greenhouse gas emissions.

California's Cap-and-Trade Program

California's program, started in 2013, covers about 80% of the state's greenhouse gas emissions and has been linked with Quebec's system since 2014.

How the Model Works

Our simulation models a simplified cap-and-trade market with 24 firms. Understanding the model requires grasping four key concepts.

The Emissions Cap

The regulator sets a total emissions cap—the maximum amount of pollution allowed across all firms.⁵ This cap is typically set below current emissions levels to force overall reductions.

The cap is the environmental guarantee: In the simulation, this is the *Emissions Cap* slider. total emissions cannot exceed this level, regardless of economic conditions.

Initial Allocation

Permits are initially distributed to firms. There are two main approaches:⁶

Grandfathering Free distribution based on past emissions

Auctioning Firms must purchase permits from the government

⁶ The simulation uses grandfathering, distributing permits equally among firms. In practice, allocation formulas are often based on historical emissions.

Firm Heterogeneity

Each firm has a unique *cost level*⁷ representing the cost of reducing emissions:

- **Low-cost firms** (0.5–0.8): Can reduce emissions cheaply through efficiency improvements or fuel switching
- **Medium-cost firms** (0.8–1.2): Face moderate abatement costs
- **High-cost firms** (1.2–1.5): Expensive to reduce emissions; may need to buy permits

The cost level determines each firm's *willingness to pay* for permits:

⁷ In the simulation, you set this via the cost level inputs, ranging from 0.5 to 1.5.

$$\text{Max WTP} = \text{Price Floor} + (\text{Price Ceiling} - \text{Price Floor}) \times \text{Cost Level} \quad (1)$$

Market Trading

After initial allocation, firms compare their allocation to their desired emissions:⁸

Surplus firms Allocated more than needed → **Sellers**

Deficit firms Allocated less than needed → **Buyers**

Trading continues until the market reaches equilibrium, where supply equals demand at a market price that clears the market.

⁸ Desired emissions reflect the firm's production needs and abatement opportunities.

A firm with surplus permits can profit by selling to deficit firms, while buyers avoid costly emission reductions.

Understanding Market Dynamics

Supply and Demand Curves

The simulation displays supply and demand curves:⁹

- **Supply curve:** Upward sloping, showing that higher prices induce more permit sales (firms reduce emissions more)
- **Demand curve:** Downward sloping, showing that higher prices reduce permit demand (firms reduce emissions to avoid buying)
- **Equilibrium:** Where curves intersect, determining market price and quantity traded

⁹ These curves are stylized representations. In reality, they would be step functions reflecting individual firm decisions.

Price Floor and Ceiling

Real markets often include price constraints:¹⁰

Price Floor Prevents permits from becoming too cheap, maintaining abatement incentives

Price Ceiling Prevents excessive costs that could harm the economy

¹⁰ California's cap-and-trade has both a price floor (auction reserve price) and a ceiling (allowance price containment reserve).

Using the Simulation

Basic Exploration

Step 1: Observe the baseline

- Start with default parameters
- Note the equilibrium price and quantity
- Examine which firms are buyers vs. sellers

Step 2: Adjust the cap

- Reduce the emissions cap from 100 to 80 tons

- What happens to the price?
- Which firms can no longer afford to trade?

Step 3: Explore firm heterogeneity

- Set Firms 1–8 to cost level 1.5 (high cost)
- Set Firms 9–16 to cost level 0.5 (low cost)
- Set Firms 17–24 to cost level 1.0 (medium cost)
- Observe trading patterns

Advanced Scenarios

Scenario 1: Stringent Cap

Setup:

- Emissions Cap: 70 tons
- Total Demand: 140 tons
- Price Floor: \$20
- Price Ceiling: \$150

Questions to explore:

1. What is the shortage of permits?
2. How many firms cannot afford to trade?
3. What is the equilibrium price?
4. How does this compare to a less stringent cap?

Scenario 2: Technology Shock

Imagine a breakthrough in clean technology reduces costs for half the firms.

Setup:

- Set Firms 1–12 to cost level 0.6 (new technology)
- Keep Firms 13–24 at their original levels

Observe:

- How does equilibrium price change?
- Do low-cost firms become sellers?
- What happens to market value?

A stringent cap creates scarcity, driving up prices and forcing more emission reductions.

Scenario 3: Price Controls

Test the impact of regulatory price constraints.

Setup A: Binding floor

- Set Price Floor to \$40 (high)
- Observe which trades don't occur

Setup B: Binding ceiling

- Set Price Ceiling to \$60 (low)
- Emissions Cap: 70 tons
- Demand: 140 tons
- What happens when demand exceeds supply at the ceiling price?

Key Economic Concepts

Equimarginal Principle

Tradable permits achieve the *equimarginal principle*:¹¹ all firms face the same marginal cost of reducing emissions (the permit price).

- Firms with low abatement costs reduce emissions and sell permits
- Firms with high abatement costs buy permits instead of reducing
- This achieves the cap at minimum total cost

¹¹ At the optimum, the marginal cost of abatement should be equal across all firms.

Static vs. Dynamic Efficiency

Static efficiency: Achieving today's emissions target at lowest cost

Dynamic efficiency: Creating incentives for long-term innovation

The simulation focuses on static efficiency, but real markets also drive innovation through the ongoing incentive to reduce costs.

Permit systems encourage firms to develop cleaner technologies. A firm that innovates can sell more permits or buy fewer.

Market Power Concerns

In real markets with few participants, firms might manipulate prices:¹²

- **Banking:** Saving permits for future use
- **Hoarding:** Withholding permits to drive up prices
- **Strategic allocation:** Gaming initial distribution

¹² The simulation assumes perfect competition, but real markets may have strategic behavior.

Extensions and Limitations

What the Model Includes

- Heterogeneous firms with different abatement costs
- Initial allocation and trading
- Price constraints
- Market equilibrium

What the Model Omits

- **Banking and borrowing:** Firms cannot save permits for future periods
- **Uncertainty:** Firms have perfect information
- **Transaction costs:** Trading is costless
- **Monitoring and enforcement:** Compliance is assumed
- **Offsets:** No credits from external reduction projects
- **International linkages:** No connection to other markets

Discussion Questions

1. Why might a firm prefer grandfathered permits over auctioned permits? What are the equity implications?
2. If you were a regulator, how would you decide between a cap-and-trade system and a carbon tax?
3. What happens to the equilibrium price as the cap becomes more stringent over time? How would firms respond?
4. Why is it important that permits be tradable? What would happen if each firm simply had to meet its initial allocation?
5. How do price floors and ceilings affect market efficiency? When might they be desirable?
6. What role does initial allocation play in determining the final distribution of emissions? The final distribution of costs?
7. How might banking (saving permits for future use) affect firm behavior and market prices?
8. Consider environmental justice: Does a cap-and-trade system guarantee that pollution will be reduced in every location? Why or why not?

Further Reading

- Stavins, R.N. (2008). Addressing climate change with a comprehensive US cap-and-trade system. *Oxford Review of Economic Policy*, 24(2), 298-321.
- Tietenberg, T. (2006). *Emissions Trading: Principles and Practice* (2nd ed.). Resources for the Future.
- Ellerman, A.D., Joskow, P.L., & Harrison, D. (2003). *Emissions Trading in the U.S.: Experience, Lessons, and Considerations for Greenhouse Gases*. Pew Center on Global Climate Change.
- Hahn, R.W., & Stavins, R.N. (2011). The Effect of Allowance Allocations on Cap-and-Trade System Performance. *Journal of Law and Economics*, 54(4), S267-S294.

Conclusion

Cap-and-trade systems represent a powerful application of economic theory to environmental policy. By creating a market for pollution rights, they harness private incentives to achieve public environmental goals efficiently.

The simulation allows you to explore how different firms with different costs interact in a permit market, how policy parameters affect outcomes, and why markets can achieve environmental goals more efficiently than command-and-control regulation.

As you experiment with different scenarios, think about the policy trade-offs involved: stringency vs. cost, efficiency vs. equity, simplicity vs. precision. These are the challenges facing real-world environmental policymakers today.

Remember: The goal is not just to understand how the model works, but to develop intuition about how market-based environmental policies function in the real world.